

Nr 5.

$$A = (Q, \Sigma, \delta, q_0, F)$$

$$\Sigma = \{1, 2, 3\}$$

$$F = \{ q(b, u, v) \mid u = b \}$$

1. $\delta(q_0, a) = q_1(a)$
2. $\delta(q_1(a), b) = q(b, a, b)$
3. $\delta(q(b, u, v), y) = q(b, v, y)$

$$Q = \{ q_0 \} \cup \{ q_1(a) \mid a \in \Sigma \} \cup \{ q(b, u, v) \mid b, u, v \in \Sigma \}$$

Nr.6

$$\begin{array}{lcl} S & \rightarrow & aA \\ A & \rightarrow & dB \mid bA \mid cA \\ B & \rightarrow & aC \mid bC \mid cA \\ C & \rightarrow & \epsilon \end{array}$$

Regex:

$$a(b|c)^*d(c(b|c)^*d)^*(a|b)$$